

Archana Senthil

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Executive Summary

- Data analytics professional with expertise in Power BI (Certified), SQL, Snowflake (Certified), and Python. Passionate about leveraging AI and automation in data workflows to drive efficiency and insights.
- Over four years of experience in data analysis, with a focus on machine learning and generative models.
- Skilled in deep learning, autoencoders, and dimensionality reduction techniques, particularly for applications like image compression and data compression.
- Versatile across sectors, with strong capabilities in data quality management, project leadership, and continuous process optimization.
- Quick to adopt new technologies and drive process improvements, with strong communication and collaboration skills that align with organizational goals.

Technical Skills

Applications & Tools: SAP (PM Module), Microsoft Power BI (Certified), Snowflake (Certified), Microsoft Excel, Microsoft Azure, Android Studio, MS Office Suite, R Studio, Microsoft SQL Server, Jupyter Notebook, SQLite.

Data Analysis/Reporting: Power BI, Excel, Python, SQL, Looker Studio, IBM Cognos Analytics.

Project Management: Agile, Waterfall, Scrum, Jira, Kanban, Trello, Gantt Charts, Asana.

Programming Languages: Python, SQL, Kotlin, R, M Query, DAX.

Machine Learning: Autoencoders, Generative Models, Neural Networks, Supervised/Unsupervised Learning

Education

Master of Data Science and Analytics

Expected Sep 2025

University of Calgary | Alberta, Canada

Projects

- **Olympic Medal Forecasting:** Developed a predictive regression model for Olympic medal forecasting for the Paris Olympics using Python in a sports analysis group project.
- **City Of Calgary Survey Analysis:** Analyzed the City of Calgary's fall survey using R, leveraging data visualization tools for effective communication of results.
- **Superconductor Temperature Prediction:** Predicted critical temperatures of superconductors using multivariate regression techniques in R.
- **LinkedIn Job Search & Mental Health Survey:** Designed and analyzed a survey to document challenges faced by MDSA students during job searches, using statistical analysis and Python.
- **Alberta Wildfire Dataset:** Implemented machine learning classification techniques and visualizations using Python, PowerBI, and Tableau, creating dashboards to analyze the impact of human activity on Alberta Wildfires. The project also included the application of generative models and dimensionality reduction techniques, particularly using autoencoders to analyze large datasets and extract features for predictive modeling.
- **PowerBI Dashboard Development:** Led the development of a Power BI dashboard, incorporating interactive Key Performance Metrics (KPI) tracking of a fictional company. Designed KPI tiles for Profit, Active Products, and Average Pay Rate, incorporating charts and semicircle donut visualizations for data representation. Held knowledge-sharing sessions on Power BI for team members, improving PowerBI dashboard adoption to support this assignment.

Research

- Supporting a professor in using AI-driven co-clustering techniques to build a model and predict research paper citations and relevancy.

Bachelor of Electrical and Electronics Engineering (WES, Ref #4321307IMM)

May 2017

Sri Venkateswara College of Engineering | Sriperumbudur, India

Certifications & Trainings

Tableau Certified Data Analyst	Expected Apr 2025
Big Query for Data Analysis	Nov 2024
Snowflake Snowpro Core Certificate	Mar 2023
Google Project Management Certificate	June 2022
Microsoft Power BI Data Analyst Associate	Apr 2022
IBM Data Analyst Professional Certificate	Sep 2021
Microsoft Azure Fundamentals Professional Certificate	Jul 2021
Microsoft Excel - Excel from beginner to Advanced	Apr 2021
The Complete SQL Bootcamp 2021: Go from Zero to Hero	Apr 2021

Work Experience

Peer Mentor – Family to Family Connections (Volunteer)	Oct 2023- Present
Alberta Health Services Calgary, Canada	
- Organized and led critical support sessions for parents in the NICU, fostering a space for open discussion and collaborative problem-solving, skills directly transferable to team-based data science projects. - Delivered empathetic support and research-backed resources to families, enhancing user-focused analysis and stakeholder communication. - Guided families in navigating complex decisions, mirroring the ability to analyze data and provide actionable insights.	
Online Data Analyst (Freelance)	Jul 2022 – Nov 2023
Telus International Calgary, Canada	
- Executed data analysis and ensured data quality by cleaning and organizing dataset for actionable insights. - Managed data quality through de-duplication and visualizations to support business goals. - Assessed AI-generated results, delivering feedback to enhance quality control and database modeling.	
Mapping and Location Coordinator (Month Contract)	Nov 2021
SHAW Communications Calgary, Canada	
- Worked with cross-functional teams to enforce data quality standards and ensure data accuracy in line with organizational and industry regulations using Civic and GIS tools. - Cleaned and organized large datasets in Excel and Updated CBS (Central Billing System), collaborating with GIS teams to enrich insights and ensuring reliable datasets for GeoSpatial Analysis.	
Software Development Intern	Sep 2021 – Oct 2021
GAO Tek Inc. Calgary, Canada	
- Developed a proficient working knowledge of the Kotlin Programming Language and Android Studio. - Scheduled daily stand-up meetings to track the application's progress using Jira and Kanban approaches. - Helped design an app and bug testing using Kotlin language supporting Bluetooth services and frameworks.	
Assistant Manager	Aug 2017 - May 2019
BALCO, Vedanta Ltd. Korba, India	
- Led data quality management practices in equipment maintenance and inventory systems, ensuring accurate, actionable reporting for decision-making. - Streamlined processes with SAP and predictive modeling, minimizing downtime and improving operational efficiency. - Conducted root cause analyses and led Lean/Six Sigma projects, enhancing equipment reliability, and safety metrics. - Drove data stewardship, ensuring proper data classification, access controls, audits, and compliance with governance. - Generated performance and expense reports using advanced Excel features, including pivot tables, to support data-driven decision-making. - Designed process diagrams in MS Visio and led kaizen initiatives, achieving a 5% improvement in shop floor safety and operational profitability.	